

Theory of Commutative Algebra 15

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Lecture notes with worked examples in commutative algebra

Contents

1 Resolutions and free resolutions: concrete examples over $\mathbb{Z}$, polynomial rings, quotients, and localizations

Convention. Throughout, A denotes a commutative ring with 1, and all modules are unital A -modules.

1.1 1. What is a resolution?

Let M be an A -module. A *resolution* of M is an exact sequence

$$\cdots \longrightarrow F_2 \longrightarrow F_1 \longrightarrow F_0 \longrightarrow M \longrightarrow 0.$$

If each F_i is a free A -module, then this is called a *free resolution* of M .

The exactness means that at every stage the image of one map equals the kernel of the next. Conceptually, a free resolution records:

- generators of M ,
- relations among those generators,
- relations among those relations,
- and so on.

Thus:

- F_0 gives generators of M ,
- F_1 gives relations among generators,
- F_2 gives relations among relations,
- etc.

1.2 2. The most basic resolution: quotients by ideals

If $I \subseteq A$ is an ideal, then there is always an exact sequence

$$0 \longrightarrow I \longrightarrow A \longrightarrow A/I \longrightarrow 0.$$

Here the map $A \rightarrow A/I$ is the quotient map, and its kernel is precisely I .

This is always a resolution of A/I , but it is not always a *free* resolution, because I need not be a free A -module.

So the sequence

$$0 \longrightarrow I \longrightarrow A \longrightarrow A/I \longrightarrow 0$$

is a resolution in general, and it becomes a free resolution only when I is free.

1.3 3. The prototype over $\mathbb{Z}$

Take $A = \mathbb{Z}$ and $M = \mathbb{Z}/n\mathbb{Z}$. Then there is an exact sequence

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\cdot n} \mathbb{Z} \longrightarrow \mathbb{Z}/n\mathbb{Z} \longrightarrow 0.$$

Let us check exactness:

- the map $\mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$ is reduction modulo n ,
- its kernel is $n\mathbb{Z}$,
- multiplication by n identifies $\mathbb{Z}$ with $n\mathbb{Z}$.

So this is a free resolution of $\mathbb{Z}/n\mathbb{Z}$ of length 1.

This is the standard model for a cyclic module over a principal ideal domain.

1.4 4. Principal quotients over $k[x]$

Let $A = k[x]$, where k is a field.

Example 1: $A/(x)$

We have an exact sequence

$$0 \longrightarrow A \xrightarrow{\cdot x} A \longrightarrow A/(x) \longrightarrow 0.$$

Since $A/(x) \cong k$, this can also be written as

$$0 \longrightarrow k[x] \xrightarrow{\cdot x} k[x] \longrightarrow k \longrightarrow 0.$$

So k has a free resolution of length 1 as a $k[x]$ -module.

Example 2: $A/(f)$

More generally, for any nonzero polynomial $f \in k[x]$,

$$0 \longrightarrow A \xrightarrow{\cdot f} A \longrightarrow A/(f) \longrightarrow 0$$

is exact.

Indeed:

- the map $A \rightarrow A/(f)$ is the quotient map,
- its kernel is (f) ,
- the principal ideal (f) is isomorphic to A via multiplication by f .

So every principal quotient of $k[x]$ has a free resolution of length 1.

1.5 5. A first higher example over $k[x, y]$

Now let $A = k[x, y]$. Consider

$$M = A/(x, y) \cong k.$$

A free resolution of k over A is

$$0 \longrightarrow A \xrightarrow{\varphi_2} A^2 \xrightarrow{\varphi_1} A \longrightarrow k \longrightarrow 0,$$

where

$$\varphi_1(a, b) = ax + by$$

and

$$\varphi_2(c) = (-cy, cx).$$

In matrix form,

$$0 \longrightarrow A \xrightarrow{\begin{pmatrix} -y \\ x \end{pmatrix}} A^2 \xrightarrow{\begin{pmatrix} x & y \end{pmatrix}} A \longrightarrow k \longrightarrow 0.$$

Let us interpret this:

- $A \rightarrow k$ gives one generator of k as an A -module,
- $A^2 \rightarrow A$ records the two relations $x = 0$ and $y = 0$ in the quotient,
- $A \rightarrow A^2$ records the relation among those relations, namely

$$x \cdot y - y \cdot x = 0.$$

This is the beginning of the Koszul resolution.

So here we see clearly why higher free modules $F_2, F_3, \dots$ appear: they capture higher-order dependencies.

1.6 6. The general pattern over $k[x_1, \dots, x_n]$

Let

$$A = k[x_1, \dots, x_n]$$

and consider

$$M = A/(x_1, \dots, x_n) \cong k.$$

Then k has a free resolution given by the Koszul complex:

$$0 \longrightarrow A^{\binom{n}{n}} \longrightarrow A^{\binom{n}{n-1}} \longrightarrow \dots \longrightarrow A^{\binom{n}{1}} \longrightarrow A \longrightarrow k \longrightarrow 0.$$

The important point is not only the exact shape of all maps, but the fact that:

- the resolution is finite,
- its length is n ,
- and it reflects the n generators $x_1, \dots, x_n$ of the maximal ideal $(x_1, \dots, x_n)$.

Thus over a polynomial ring in n variables, the residue field at the origin has a free resolution of length n .

1.7 7. Over a field: every module is already free

If $A = K$ is a field, then every A -module is just a vector space over K . Every vector space has a basis, so every module is free.

Therefore the theory becomes trivial: if M is a K -vector space, then a free resolution is simply

$$0 \longrightarrow M \xrightarrow{\text{id}} M \longrightarrow 0,$$

or even one may say that no nontrivial resolution is needed, because M is already free.

So over a field, free resolutions exist but are usually uninteresting.

1.8 8. The case $\mathbb{Z}/p\mathbb{Z}$

If by $\mathbb{Z}_p$ one means $\mathbb{Z}/p\mathbb{Z}$, then this ring is a field, usually denoted $\mathbb{F}_p$.

Hence every $\mathbb{F}_p$ -module is a vector space, so every module is free. Therefore all free resolutions over $\mathbb{F}_p$ are trivial in the same sense as over any field.

For example, if $M = \mathbb{F}_p^m$, then M is already a free $\mathbb{F}_p$ -module of rank m .

1.9 9. The localized ring $\mathbb{Z}_{(p)}$

Sometimes one writes $\mathbb{Z}_p$ informally when one actually means the localization

$$\mathbb{Z}_{(p)} = \left\{ \frac{a}{b} \in \mathbb{Q} : p \nmid b \right\}.$$

This is the localization of $\mathbb{Z}$ at the prime ideal (p) . It is a local principal ideal domain.

Then we have the exact sequence

$$0 \longrightarrow \mathbb{Z}_{(p)} \xrightarrow{p} \mathbb{Z}_{(p)} \longrightarrow \mathbb{Z}_{(p)}/(p) \longrightarrow 0.$$

Since

$$\mathbb{Z}_{(p)}/(p) \cong \mathbb{F}_p,$$

this gives a free resolution of $\mathbb{F}_p$ as a $\mathbb{Z}_{(p)}$ -module.

So this is the localized analogue of

$$0 \longrightarrow \mathbb{Z} \xrightarrow{p} \mathbb{Z} \longrightarrow \mathbb{Z}/p\mathbb{Z} \longrightarrow 0.$$

1.10 10. The p -adic integers $\mathbb{Z}_p$

If instead $\mathbb{Z}_p$ means the ring of p -adic integers, then similarly

$$0 \longrightarrow \mathbb{Z}_p \xrightarrow{p} \mathbb{Z}_p \longrightarrow \mathbb{Z}_p/(p) \longrightarrow 0$$

is exact, and

$$\mathbb{Z}_p/(p) \cong \mathbb{F}_p.$$

So $\mathbb{F}_p$ again has a free resolution of length 1 over $\mathbb{Z}_p$.

1.11 11. Free resolutions over a general ring

Let A be any ring and M any A -module.

A free resolution always exists. Indeed:

- choose a surjection $F_0 \twoheadrightarrow M$ from a free module F_0 ,
- let $K_1 = \ker(F_0 \rightarrow M)$,
- choose a surjection $F_1 \twoheadrightarrow K_1$ from a free module F_1 ,
- let $K_2 = \ker(F_1 \rightarrow F_0)$,
- and continue inductively.

Thus every module admits a free resolution

$$\cdots \longrightarrow F_2 \longrightarrow F_1 \longrightarrow F_0 \longrightarrow M \longrightarrow 0.$$

However, unlike the PID or polynomial examples above, such a resolution may:

- be very long,
- be infinite,
- be difficult to describe explicitly,
- and be highly non-minimal.

1.12 12. What happens under quotient?

Suppose $I \subseteq A$ is an ideal and $B = A/I$.

If

$$\cdots \longrightarrow F_2 \longrightarrow F_1 \longrightarrow F_0 \longrightarrow M \longrightarrow 0$$

is a free resolution of an A -module M , then tensoring with A/I gives a complex

$$\cdots \longrightarrow F_2 \otimes_A A/I \longrightarrow F_1 \otimes_A A/I \longrightarrow F_0 \otimes_A A/I \longrightarrow M \otimes_A A/I \longrightarrow 0.$$

Since each F_i is free, each tensor product $F_i \otimes_A A/I$ is a free A/I -module:

$$F_i \cong A^{r_i} \quad \implies \quad F_i \otimes_A A/I \cong (A/I)^{r_i}.$$

So quotienting produces a new complex of free A/I -modules. But the crucial point is:

Tensoring with A/I is right exact, but not always exact.

Therefore the new complex need not remain exact on the left. So a free resolution over A does *not* in general stay a free resolution after quotienting.

Concrete example

Start with the free resolution over $\mathbb{Z}$:

$$0 \longrightarrow \mathbb{Z} \xrightarrow{2} \mathbb{Z} \longrightarrow \mathbb{Z}/2\mathbb{Z} \longrightarrow 0.$$

Tensor with $\mathbb{Z}/2\mathbb{Z}$. Then multiplication by 2 becomes the zero map, so we get

$$0 \longrightarrow \mathbb{Z}/2\mathbb{Z} \xrightarrow{0} \mathbb{Z}/2\mathbb{Z} \longrightarrow (\mathbb{Z}/2\mathbb{Z}) \otimes_{\mathbb{Z}} (\mathbb{Z}/2\mathbb{Z}) \longrightarrow 0.$$

This is no longer exact at the left-hand copy of $\mathbb{Z}/2\mathbb{Z}$. So exactness can fail after quotienting.

This is one of the reasons Tor appears in homological algebra.

1.13 13. What happens under localization?

Localization behaves much better.

Let $S \subseteq A$ be a multiplicative set. If

$$\cdots \longrightarrow F_2 \longrightarrow F_1 \longrightarrow F_0 \longrightarrow M \longrightarrow 0$$

is a free resolution of M , then localizing gives

$$\cdots \longrightarrow S^{-1}F_2 \longrightarrow S^{-1}F_1 \longrightarrow S^{-1}F_0 \longrightarrow S^{-1}M \longrightarrow 0.$$

Since localization is an exact functor, this sequence remains exact.

Also, if $F_i \cong A^{r_i}$, then

$$S^{-1}F_i \cong (S^{-1}A)^{r_i},$$

so free modules remain free after localization.

Therefore:

Localization of a free resolution is again a free resolution.

Concrete example

Start with

$$0 \longrightarrow \mathbb{Z} \xrightarrow{n} \mathbb{Z} \longrightarrow \mathbb{Z}/n\mathbb{Z} \longrightarrow 0.$$

Now localize at $S = \mathbb{Z} \setminus (p)$. Then we obtain

$$0 \longrightarrow \mathbb{Z}_{(p)} \xrightarrow{n} \mathbb{Z}_{(p)} \longrightarrow (\mathbb{Z}/n\mathbb{Z})_{(p)} \longrightarrow 0.$$

There are two cases:

- if $p \nmid n$, then n becomes a unit in $\mathbb{Z}_{(p)}$, so multiplication by n is an isomorphism and

$$(\mathbb{Z}/n\mathbb{Z})_{(p)} = 0;$$

- if $p \mid n$, then some p -torsion remains, so the localization is nonzero.

Thus localization isolates the part of the module that lives near the prime (p) .

1.14 14. Why free resolutions matter

Free resolutions are important because they allow us to define and compute derived functors.

Tor

If $F_\bullet \rightarrow M$ is a free resolution of M , then

$$\mathrm{Tor}_i^A(M, N) = H_i(F_\bullet \otimes_A N).$$

So Tor measures the failure of tensor product to preserve exactness.

Ext

If $F_\bullet \rightarrow M$ is a free resolution of M , then

$$\mathrm{Ext}_A^i(M, N) = H^i(\mathrm{Hom}_A(F_\bullet, N)).$$

So Ext records higher extension data and higher homological information about modules.

1.15 15. Summary of the most important examples

The key examples worth remembering are:

1. Over $\mathbb{Z}$:

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\cdot n} \mathbb{Z} \longrightarrow \mathbb{Z}/n\mathbb{Z} \longrightarrow 0.$$

2. Over $k[x]$:

$$0 \longrightarrow k[x] \xrightarrow{\cdot f} k[x] \longrightarrow k[x]/(f) \longrightarrow 0.$$

3. Over $k[x, y]$:

$$0 \longrightarrow A \xrightarrow{\begin{pmatrix} -y \\ x \end{pmatrix}} A^2 \xrightarrow{\begin{pmatrix} x & y \end{pmatrix}} A \longrightarrow k \longrightarrow 0, \quad A = k[x, y].$$

4. Under localization: if $F_\bullet \rightarrow M$ is a free resolution over A , then

$$S^{-1}F_\bullet \rightarrow S^{-1}M$$

is again a free resolution over $S^{-1}A$.

5. Under quotient: if $F_\bullet \rightarrow M$ is a free resolution over A , then

$$F_\bullet \otimes_A A/I$$

is a complex of free A/I -modules, but it need not remain exact.

1.16 16. Final intuition

A free resolution is a systematic way of describing a module by successively recording:

$$\text{generators} \rightarrow \text{relations} \rightarrow \text{relations among relations} \rightarrow \cdots$$

This is why resolutions are central both in concrete algebra and in homological algebra.

2 Companion examples and worked exercises on free resolutions

Purpose of this companion section. The previous section introduced resolutions and free resolutions in general form. This companion section is meant to slow the discussion down and show, in concrete terms, how one actually reads, builds, and checks free resolutions over familiar rings such as $\mathbb{Z}$, polynomial rings, quotient rings, and localizations.

2.1 1. How to read a free resolution

Suppose we have a free resolution

$$\cdots \longrightarrow F_2 \xrightarrow{d_2} F_1 \xrightarrow{d_1} F_0 \xrightarrow{\varepsilon} M \longrightarrow 0.$$

The meaning of the first few terms is as follows:

- F_0 is a free module mapping onto M , so its basis elements may be viewed as *formal generators* of M .
- $\ker(\varepsilon) \subseteq F_0$ consists of the relations among those generators.
- F_1 maps onto $\ker(\varepsilon)$, so its basis elements encode *formal relations*.
- $\ker(d_1) \subseteq F_1$ consists of relations among those relations.
- F_2 maps onto $\ker(d_1)$, and the process continues.

So a free resolution is not just a sequence of modules and maps. It is a systematic record of the algebraic structure of M :

$$\text{generators} \rightarrow \text{relations} \rightarrow \text{relations among relations} \rightarrow \cdots$$

2.2 2. A very first warm-up: the module $\mathbb{Z}/6\mathbb{Z}$

Consider the $\mathbb{Z}$ -module

$$M = \mathbb{Z}/6\mathbb{Z}.$$

A free resolution is

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\cdot 6} \mathbb{Z} \longrightarrow \mathbb{Z}/6\mathbb{Z} \longrightarrow 0.$$

Why is this exact?

The map $\mathbb{Z} \rightarrow \mathbb{Z}/6\mathbb{Z}$ is the quotient map $n \mapsto \bar{n}$. Its kernel is

$$\ker(\mathbb{Z} \rightarrow \mathbb{Z}/6\mathbb{Z}) = 6\mathbb{Z}.$$

The map $\mathbb{Z} \xrightarrow{6} \mathbb{Z}$ has image $6\mathbb{Z}$, so exactness holds.

Interpretation. As a $\mathbb{Z}$ -module, M is generated by one element, namely $\bar{1}$, and there is one relation:

$$6 \cdot \bar{1} = 0.$$

Thus one free generator and one free relation are enough.

2.3 3. A polynomial example: $k[x]/(x)$

Let $A = k[x]$, where k is a field, and consider

$$M = A/(x).$$

Since $A/(x) \cong k$, we may think of this as the residue field at the point $x = 0$.

A free resolution is

$$0 \longrightarrow A \xrightarrow{x} A \longrightarrow A/(x) \longrightarrow 0.$$

Why is this exact?

The quotient map $A \rightarrow A/(x)$ has kernel (x) . But $(x) \cong A$ via multiplication by x , because

$$A \longrightarrow (x), \quad f \mapsto xf$$

is an isomorphism of A -modules.

Interpretation. The module $A/(x)$ is generated by the class of 1, and the only relation is

$$x \cdot 1 = 0 \quad \text{in } A/(x).$$

2.4 4. Another polynomial example: $k[x]/(x^2)$

Now consider

$$M = A/(x^2), \quad A = k[x].$$

Again there is a free resolution

$$0 \longrightarrow A \xrightarrow{x^2} A \longrightarrow A/(x^2) \longrightarrow 0.$$

This looks formally similar to the previous example, but the quotient is different.

Important distinction. The quotient $A/(x)$ is reduced and isomorphic to the field k , while $A/(x^2)$ is not reduced: the class of x is nonzero in $A/(x^2)$, but

$$x^2 = 0 \quad \text{in } A/(x^2).$$

So the same style of free resolution can describe very different quotient modules.

2.5 5. A first genuinely higher example: $k[x, y]/(x, y)$

Let

$$A = k[x, y], \quad M = A/(x, y) \cong k.$$

This module is generated by the class of 1, but now there are two obvious relations:

$$x \cdot 1 = 0, \quad y \cdot 1 = 0.$$

These are not independent in a completely trivial way, because there is a relation between them:

$$y(x \cdot 1) - x(y \cdot 1) = 0.$$

This is exactly what the free resolution records:

$$0 \longrightarrow A \xrightarrow{\begin{pmatrix} -y \\ x \end{pmatrix}} A^2 \xrightarrow{\begin{pmatrix} x & y \end{pmatrix}} A \longrightarrow k \longrightarrow 0.$$

Let us check that the composition is zero:

$$\begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} -y \\ x \end{pmatrix} = -xy + yx = 0.$$

Now let us describe the maps:

- The map $A \rightarrow k$ is evaluation at $(0, 0)$.
- The map $A^2 \rightarrow A$ sends $(a, b) \mapsto ax + by$.
- The map $A \rightarrow A^2$ sends $c \mapsto (-cy, cx)$.

Interpretation. This says:

- one generator for k ,
- two relations $x = 0$ and $y = 0$,
- one relation among these relations, namely $(-y, x)$.

This is the simplest nontrivial example showing why F_2 appears.

2.6 6. The same picture in $k[x_1, \dots, x_n]$

Let

$$A = k[x_1, \dots, x_n], \quad M = A/(x_1, \dots, x_n) \cong k.$$

Then M has a free resolution given by the Koszul complex. The beginning and end look like

$$0 \longrightarrow A^{\binom{n}{n}} \longrightarrow A^{\binom{n}{n-1}} \longrightarrow \dots \longrightarrow A^n \longrightarrow A \longrightarrow k \longrightarrow 0.$$

The philosophical meaning is the same as before:

- one generator of k ,
- n defining relations $x_1, \dots, x_n$,
- then higher compatibility relations among those relations,
- and all of these are encoded in a finite exact complex.

2.7 7. What changes for quotient rings?

Suppose A is a ring and $I \subseteq A$ is an ideal. Then the quotient $B = A/I$ is a new ring. A module over B is the same thing as an A -module on which every element of I acts as zero.

This point of view is often useful for resolutions.

Example: $A = k[x]$, $B = A/(x) \cong k$

As an A -module, B has the free resolution

$$0 \longrightarrow A \xrightarrow{\cdot x} A \longrightarrow B \longrightarrow 0.$$

But as a B -module, the same object is just B itself, so it is free of rank 1 over B , and the resolution collapses to the trivial one

$$0 \longrightarrow B \xrightarrow{\text{id}} B \longrightarrow 0.$$

Lesson. The shape of a resolution depends strongly on which ring we are using as the base ring.

2.8 8. What happens under localization?

Let $S \subseteq A$ be a multiplicative set. If

$$\cdots \longrightarrow F_2 \longrightarrow F_1 \longrightarrow F_0 \longrightarrow M \longrightarrow 0$$

is a free resolution over A , then

$$\cdots \longrightarrow S^{-1}F_2 \longrightarrow S^{-1}F_1 \longrightarrow S^{-1}F_0 \longrightarrow S^{-1}M \longrightarrow 0$$

is a free resolution over $S^{-1}A$.

This works because:

- localization is exact,
- and localization preserves free modules:

$$S^{-1}(A^r) \cong (S^{-1}A)^r.$$

Example: localizing $\mathbb{Z}/6\mathbb{Z}$ at (5)

Start with

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\cdot 6} \mathbb{Z} \longrightarrow \mathbb{Z}/6\mathbb{Z} \longrightarrow 0.$$

Localize at $S = \mathbb{Z} \setminus (5)$. Then

$$0 \longrightarrow \mathbb{Z}_{(5)} \xrightarrow{\cdot 6} \mathbb{Z}_{(5)} \longrightarrow (\mathbb{Z}/6\mathbb{Z})_{(5)} \longrightarrow 0.$$

Since 6 is invertible in $\mathbb{Z}_{(5)}$, multiplication by 6 is an isomorphism. Therefore

$$(\mathbb{Z}/6\mathbb{Z})_{(5)} = 0.$$

Example: localizing $\mathbb{Z}/6\mathbb{Z}$ at (2)

Now localize at $S = \mathbb{Z} \setminus (2)$:

$$0 \longrightarrow \mathbb{Z}_{(2)} \xrightarrow{\cdot 6} \mathbb{Z}_{(2)} \longrightarrow (\mathbb{Z}/6\mathbb{Z})_{(2)} \longrightarrow 0.$$

Since $6 = 2 \cdot 3$, and 3 is invertible in $\mathbb{Z}_{(2)}$, multiplication by 6 has the same image as multiplication by 2. Thus the quotient is isomorphic to

$$\mathbb{Z}_{(2)}/(2) \cong \mathbb{F}_2.$$

Lesson. Localization keeps only the part of the module visible at the chosen prime.

2.9 9. What happens under quotienting the whole complex?

Let

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\cdot 2} \mathbb{Z} \longrightarrow \mathbb{Z}/2\mathbb{Z} \longrightarrow 0$$

be the standard free resolution over $\mathbb{Z}$.

Now tensor with $\mathbb{Z}/2\mathbb{Z}$. We get

$$0 \longrightarrow \mathbb{Z}/2\mathbb{Z} \xrightarrow{0} \mathbb{Z}/2\mathbb{Z} \longrightarrow (\mathbb{Z}/2\mathbb{Z}) \otimes_{\mathbb{Z}} (\mathbb{Z}/2\mathbb{Z}) \longrightarrow 0.$$

The first map is now zero, so the complex is no longer exact at the left-hand term.

Lesson. Passing from A to A/I by tensoring with A/I does not in general preserve exactness. So free resolutions behave well under localization, but not automatically under quotient.

2.10 10. The field case and the residue field case

If K is a field, every K -module is a vector space, hence free. So over a field there is no need for an interesting resolution.

Example: K^3

As a K -module, K^3 is already free of rank 3. One may write the trivial resolution

$$0 \longrightarrow K^3 \xrightarrow{\text{id}} K^3 \longrightarrow 0.$$

Example: $\mathbb{F}_p$ over $\mathbb{F}_p$

The module $\mathbb{F}_p$ over itself is free of rank 1, so again the free resolution is trivial.

Example: $\mathbb{F}_p$ over $\mathbb{Z}$

Now the situation changes:

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\cdot p} \mathbb{Z} \longrightarrow \mathbb{F}_p \longrightarrow 0.$$

So $\mathbb{F}_p$ is not free as a $\mathbb{Z}$ -module, but it does have a short free resolution over $\mathbb{Z}$.

2.11 11. Free resolutions over A versus over $A_{\mathfrak{p}}$

Let M be an A -module, and let $\mathfrak{p} \in \operatorname{Spec}(A)$. Then localizing a free resolution of M gives a free resolution of $M_{\mathfrak{p}}$ over $A_{\mathfrak{p}}$.

This is one of the main reasons local methods are so powerful: complicated global questions can often be tested after localization.

Example: $A = k[x]$, $M = A/(x(x-1))$

Over A , the module M has free resolution

$$0 \longrightarrow A \xrightarrow{\cdot x(x-1)} A \longrightarrow M \longrightarrow 0.$$

Now localize at (x) . In $A_{(x)}$, the element $x-1$ is invertible, so multiplication by $x(x-1)$ is equivalent, up to a unit, to multiplication by x . Hence

$$M_{(x)} \cong A_{(x)}/(x).$$

If instead we localize at $(x-1)$, then x becomes invertible and

$$M_{(x-1)} \cong A_{(x-1)}/(x-1).$$

So the global quotient splits into different local pieces depending on the chosen prime.

2.12 12. A worked theorem-style summary

Proposition. Let A be a ring and M an A -module.

1. M always admits a free resolution.
2. If $S \subseteq A$ is multiplicative, then localization sends a free resolution of M to a free resolution of $S^{-1}M$.
3. If $I \subseteq A$ is an ideal, tensoring a free resolution of M with A/I gives a complex of free A/I -modules, but not necessarily an exact one.

Explanation. The first point follows by repeatedly choosing surjections from free modules onto kernels. The second follows because localization is exact and preserves freeness. The third fails in general because tensor product is only right exact.

2.13 13. Worked exercises

Exercise 1

Find a free resolution of $\mathbb{Z}/12\mathbb{Z}$ as a $\mathbb{Z}$ -module.

Solution. The standard resolution is

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\cdot 12} \mathbb{Z} \longrightarrow \mathbb{Z}/12\mathbb{Z} \longrightarrow 0.$$

The quotient map has kernel $12\mathbb{Z}$, which is the image of multiplication by 12.

Exercise 2

Find a free resolution of $k[x]/(x^3)$ as a $k[x]$ -module.

Solution. We have

$$0 \longrightarrow k[x] \xrightarrow{\cdot x^3} k[x] \longrightarrow k[x]/(x^3) \longrightarrow 0.$$

This is exact because the kernel of the quotient map is the principal ideal (x^3) , which is isomorphic to $k[x]$.

Exercise 3

Describe a free resolution of k as a $k[x, y]$ -module.

Solution. Since

$$k \cong k[x, y]/(x, y),$$

a free resolution is

$$0 \longrightarrow A \xrightarrow{\begin{pmatrix} -y \\ x \end{pmatrix}} A^2 \xrightarrow{(x \ y)} A \longrightarrow k \longrightarrow 0, \quad A = k[x, y].$$

The two relations are $x = 0$ and $y = 0$, and the relation among them is $(-y, x)$.

Exercise 4

What is the localization of $\mathbb{Z}/15\mathbb{Z}$ at (2) ?

Solution. Start with

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\cdot 15} \mathbb{Z} \longrightarrow \mathbb{Z}/15\mathbb{Z} \longrightarrow 0.$$

After localizing at (2) , the element 15 becomes invertible in $\mathbb{Z}_{(2)}$, since $2 \nmid 15$. Therefore multiplication by 15 is an isomorphism, and the quotient becomes zero:

$$(\mathbb{Z}/15\mathbb{Z})_{(2)} = 0.$$

Exercise 5

What is the localization of $\mathbb{Z}/15\mathbb{Z}$ at (3)?

Solution. In $\mathbb{Z}_{(3)}$, the element 5 is invertible, so multiplication by $15 = 3 \cdot 5$ is equivalent, up to a unit, to multiplication by 3. Therefore

$$(\mathbb{Z}/15\mathbb{Z})_{(3)} \cong \mathbb{Z}_{(3)}/(3) \cong \mathbb{F}_3.$$

Exercise 6

Explain why $k[x]/(x)$ has a trivial free resolution over $k[x]/(x)$, but a nontrivial free resolution over $k[x]$.

Solution. As a module over $k[x]/(x) \cong k$, it is simply k , which is free of rank 1, so the resolution is trivial. As a module over $k[x]$, it is $k[x]/(x)$, which is not free over $k[x]$, but it has the nontrivial free resolution

$$0 \longrightarrow k[x] \xrightarrow{x} k[x] \longrightarrow k[x]/(x) \longrightarrow 0.$$

2.14 14. A compact comparison table

Base ring	Module	Typical free resolution
$\mathbb{Z}$	$\mathbb{Z}/n\mathbb{Z}$	$0 \rightarrow \mathbb{Z} \xrightarrow{n} \mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z} \rightarrow 0$
$k[x]$	$k[x]/(f)$	$0 \rightarrow k[x] \xrightarrow{f} k[x] \rightarrow k[x]/(f) \rightarrow 0$
$k[x, y]$	$k[x, y]/(x, y) \cong k$	$0 \rightarrow A \rightarrow A^2 \rightarrow A \rightarrow k \rightarrow 0$
$\mathbb{F}_p$	$\mathbb{F}_p^m$	already free
$A_{\mathfrak{p}}$	$M_{\mathfrak{p}}$	localize a free resolution of M
A/I	$M \otimes_A A/I$	tensor the complex; exactness may fail

2.15 15. Final perspective

Free resolutions are one of the main ways algebra keeps track of hidden structure. At first glance, a module may look simple because it is defined by generators and relations. But the relations themselves may satisfy further relations, and those may satisfy still more. A free resolution is the device that records this whole hierarchy.

For principal quotients such as $\mathbb{Z}/n\mathbb{Z}$ or $k[x]/(f)$, the structure stops very quickly. For quotients such as $k[x, y]/(x, y)$, a higher step already appears. For more complicated rings and modules, the hierarchy can continue for a long time, and this is precisely why resolutions become central in homological algebra.

3 Step-by-step analysis of maps, images, and kernels in the basic resolution examples

Purpose. In the previous sections, we wrote down several free resolutions and related complexes. Here we go through the main examples carefully, and for each one we identify:

the maps, $\ker(\text{map})$, $\text{im}(\text{map})$,

and we check exactness step by step.

3.1 1. The basic example over $\mathbb{Z}$: $\mathbb{Z}/n\mathbb{Z}$

Consider the sequence

$$0 \longrightarrow \mathbb{Z} \xrightarrow{d_1} \mathbb{Z} \xrightarrow{d_0} \mathbb{Z}/n\mathbb{Z} \longrightarrow 0,$$

where

$$d_1(m) = nm \quad \text{and} \quad d_0(a) = \bar{a}.$$

Step 1: the map $d_1 : \mathbb{Z} \rightarrow \mathbb{Z}$

This is multiplication by n :

$$d_1(m) = nm.$$

Its image is

$$\text{im}(d_1) = n\mathbb{Z}.$$

Its kernel is

$$\ker(d_1) = \{m \in \mathbb{Z} : nm = 0\}.$$

Since $\mathbb{Z}$ has no zero divisors and $n \neq 0$, this gives

$$\ker(d_1) = 0.$$

So exactness at the first copy of $\mathbb{Z}$ is:

$$\text{im}(0 \rightarrow \mathbb{Z}) = 0 = \ker(d_1).$$

Step 2: the map $d_0 : \mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$

This is the quotient map:

$$d_0(a) = \bar{a}.$$

Its image is all of $\mathbb{Z}/n\mathbb{Z}$, since every residue class is $\bar{a}$ for some $a \in \mathbb{Z}$. Hence

$$\text{im}(d_0) = \mathbb{Z}/n\mathbb{Z}.$$

Its kernel is

$$\ker(d_0) = \{a \in \mathbb{Z} : \bar{a} = 0\}.$$

This means precisely that a is divisible by n , so

$$\ker(d_0) = n\mathbb{Z}.$$

But we already computed

$$\operatorname{im}(d_1) = n\mathbb{Z}.$$

Thus

$$\operatorname{im}(d_1) = \ker(d_0).$$

Conclusion

The sequence

$$0 \longrightarrow \mathbb{Z} \xrightarrow{n} \mathbb{Z} \longrightarrow \mathbb{Z}/n\mathbb{Z} \longrightarrow 0$$

is exact.

3.2 2. The principal quotient over $k[x]$: $k[x]/(x)$

Let $A = k[x]$, and consider

$$0 \longrightarrow A \xrightarrow{d_1} A \xrightarrow{d_0} A/(x) \longrightarrow 0,$$

where

$$d_1(f) = xf \quad \text{and} \quad d_0(g) = g + (x).$$

Step 1: the map $d_1 : A \rightarrow A$

This is multiplication by x :

$$d_1(f) = xf.$$

Its image is

$$\operatorname{im}(d_1) = \{xf : f \in A\} = (x).$$

Its kernel is

$$\ker(d_1) = \{f \in A : xf = 0\}.$$

Since $k[x]$ is an integral domain and $x \neq 0$, this implies

$$\ker(d_1) = 0.$$

Step 2: the map $d_0 : A \rightarrow A/(x)$

This is the quotient map:

$$d_0(g) = g + (x).$$

Its image is all of $A/(x)$, because every coset has the form $g + (x)$. Hence

$$\operatorname{im}(d_0) = A/(x).$$

Its kernel is

$$\ker(d_0) = \{g \in A : g + (x) = (x)\} = (x).$$

But

$$\operatorname{im}(d_1) = (x),$$

so

$$\operatorname{im}(d_1) = \ker(d_0).$$

Conclusion

The sequence

$$0 \longrightarrow k[x] \xrightarrow{x} k[x] \longrightarrow k[x]/(x) \longrightarrow 0$$

is exact.

Since $k[x]/(x) \cong k$, this is also a free resolution of k as a $k[x]$ -module.

3.3 3. The general principal quotient over $k[x]$: $k[x]/(f)$

Now let $f \in k[x]$ be nonzero, and consider

$$0 \longrightarrow A \xrightarrow{d_1} A \xrightarrow{d_0} A/(f) \longrightarrow 0,$$

where

$$d_1(h) = fh \quad \text{and} \quad d_0(g) = g + (f).$$

Step 1: the map $d_1 : A \rightarrow A$

This is multiplication by f .

Its image is

$$\operatorname{im}(d_1) = \{fh : h \in A\} = (f).$$

Its kernel is

$$\ker(d_1) = \{h \in A : fh = 0\}.$$

Since $A = k[x]$ is an integral domain and $f \neq 0$, we get

$$\ker(d_1) = 0.$$

Step 2: the map $d_0 : A \rightarrow A/(f)$

This is the quotient map.

Its image is

$$\operatorname{im}(d_0) = A/(f),$$

because every coset comes from some polynomial.

Its kernel is

$$\ker(d_0) = (f).$$

Since

$$\operatorname{im}(d_1) = (f),$$

we obtain

$$\operatorname{im}(d_1) = \ker(d_0).$$

Conclusion

The sequence

$$0 \longrightarrow k[x] \xrightarrow{f} k[x] \longrightarrow k[x]/(f) \longrightarrow 0$$

is exact.

3.4 4. The first higher example: $k[x, y]/(x, y) \cong k$

Let

$$A = k[x, y].$$

Consider the complex

$$0 \longrightarrow A \xrightarrow{d_2} A^2 \xrightarrow{d_1} A \xrightarrow{d_0} k \longrightarrow 0,$$

where

$$d_2(c) = (-cy, cx),$$

$$d_1(a, b) = ax + by,$$

and

$$d_0(f) = f(0, 0).$$

We now check exactness step by step.

Step 1: the map $d_0 : A \rightarrow k$

This is evaluation at the origin:

$$d_0(f) = f(0, 0).$$

Its image is all of k , because any scalar $\lambda \in k$ is obtained from the constant polynomial λ . Thus

$$\operatorname{im}(d_0) = k.$$

Its kernel is the set of all polynomials vanishing at $(0, 0)$. That is exactly the ideal

$$\ker(d_0) = (x, y).$$

So

$$\ker(d_0) = \{f \in A : f(0, 0) = 0\} = (x, y).$$

Step 2: the map $d_1 : A^2 \rightarrow A$

This is

$$d_1(a, b) = ax + by.$$

Its image is

$$\text{im}(d_1) = \{ax + by : a, b \in A\} = (x, y).$$

Since we already computed

$$\ker(d_0) = (x, y),$$

it follows that

$$\text{im}(d_1) = \ker(d_0).$$

So exactness holds at A .

Step 3: the map $d_2 : A \rightarrow A^2$

This is

$$d_2(c) = (-cy, cx).$$

Its image is

$$\text{im}(d_2) = \{(-cy, cx) : c \in A\}.$$

Now we compute $\ker(d_1)$. An element $(a, b) \in A^2$ lies in $\ker(d_1)$ if and only if

$$d_1(a, b) = ax + by = 0.$$

We claim that every such pair has the form

$$(a, b) = (-cy, cx)$$

for some $c \in A$. Hence

$$\ker(d_1) = \text{im}(d_2).$$

Why? From

$$ax = -by$$

we see that y divides ax . Since x and y are relatively prime in $k[x, y]$, this forces $y \mid a$. So write

$$a = -cy$$

for some $c \in A$. Substituting into $ax + by = 0$, we get

$$(-cy)x + by = 0,$$

so

$$y(-cx + b) = 0.$$

Since A is a domain, this implies

$$b = cx.$$

Therefore

$$(a, b) = (-cy, cx) = d_2(c).$$

Thus

$$\ker(d_1) = \text{im}(d_2).$$

Step 4: kernel of d_2

Now compute

$$\ker(d_2) = \{c \in A : (-cy, cx) = (0, 0)\}.$$

This means

$$cy = 0 \quad \text{and} \quad cx = 0.$$

Since $A = k[x, y]$ is a domain and $x, y \neq 0$, we get

$$c = 0.$$

Hence

$$\ker(d_2) = 0.$$

Conclusion

The sequence

$$0 \longrightarrow A \xrightarrow{\begin{pmatrix} -y \\ x \end{pmatrix}} A^2 \xrightarrow{\begin{pmatrix} x & y \end{pmatrix}} A \xrightarrow{f \mapsto f(0,0)} k \longrightarrow 0$$

is exact.

3.5 5. A quotient-ring viewpoint: $k[x]/(x)$ over two different base rings

This is not a new complex, but it is a very important comparison.

Let

$$A = k[x], \quad B = A/(x) \cong k.$$

As an A -module

The module B has the exact sequence

$$0 \longrightarrow A \xrightarrow{\cdot x} A \longrightarrow B \longrightarrow 0.$$

We already checked:

$$\operatorname{im}(\cdot x) = (x), \quad \ker(A \rightarrow B) = (x).$$

So over A , the module B has a nontrivial free resolution.

As a B -module

Now regard B as a module over itself. Then the relevant map is simply

$$\text{id}_B : B \rightarrow B.$$

Its image is

$$\text{im}(\text{id}_B) = B,$$

and its kernel is

$$\ker(\text{id}_B) = 0.$$

So the trivial exact sequence is

$$0 \longrightarrow B \xrightarrow{\text{id}} B \longrightarrow 0.$$

This shows that the same object can have very different free resolutions depending on the base ring.

3.6 6. Localization example: $(\mathbb{Z}/6\mathbb{Z})_{(5)}$

Start with

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\cdot 6} \mathbb{Z} \longrightarrow \mathbb{Z}/6\mathbb{Z} \longrightarrow 0.$$

Now localize at

$$S = \mathbb{Z} \setminus (5).$$

Then we obtain

$$0 \longrightarrow \mathbb{Z}_{(5)} \xrightarrow{d_1} \mathbb{Z}_{(5)} \xrightarrow{d_0} (\mathbb{Z}/6\mathbb{Z})_{(5)} \longrightarrow 0,$$

where

$$d_1(r) = 6r.$$

Since $5 \nmid 6$, the element 6 is a unit in $\mathbb{Z}_{(5)}$. Therefore multiplication by 6 is an isomorphism.

So:

$$\text{im}(d_1) = \mathbb{Z}_{(5)}, \quad \ker(d_1) = 0.$$

Because d_1 is surjective, the quotient is zero:

$$(\mathbb{Z}/6\mathbb{Z})_{(5)} = 0.$$

Thus d_0 is the zero map

$$d_0 : \mathbb{Z}_{(5)} \rightarrow 0,$$

whose image is

$$\text{im}(d_0) = 0$$

and whose kernel is

$$\ker(d_0) = \mathbb{Z}_{(5)}.$$

This matches exactness, because

$$\text{im}(d_1) = \mathbb{Z}_{(5)} = \ker(d_0).$$

3.7 7. Localization example: $(\mathbb{Z}/6\mathbb{Z})_{(2)}$

Now localize at

$$S = \mathbb{Z} \setminus (2).$$

Then

$$0 \longrightarrow \mathbb{Z}_{(2)} \xrightarrow{d_1} \mathbb{Z}_{(2)} \xrightarrow{d_0} (\mathbb{Z}/6\mathbb{Z})_{(2)} \longrightarrow 0,$$

with

$$d_1(r) = 6r.$$

Now $6 = 2 \cdot 3$, and 3 is a unit in $\mathbb{Z}_{(2)}$. So multiplication by 6 has the same image as multiplication by 2. Hence

$$\text{im}(d_1) = 6\mathbb{Z}_{(2)} = 2\mathbb{Z}_{(2)}.$$

Its kernel is still

$$\ker(d_1) = 0,$$

because $\mathbb{Z}_{(2)}$ is a domain.

The quotient is

$$(\mathbb{Z}/6\mathbb{Z})_{(2)} \cong \mathbb{Z}_{(2)}/2\mathbb{Z}_{(2)} \cong \mathbb{F}_2.$$

The map d_0 is the quotient map

$$d_0 : \mathbb{Z}_{(2)} \rightarrow \mathbb{Z}_{(2)}/2\mathbb{Z}_{(2)}.$$

So

$$\text{im}(d_0) = \mathbb{Z}_{(2)}/2\mathbb{Z}_{(2)},$$

and

$$\ker(d_0) = 2\mathbb{Z}_{(2)}.$$

Since

$$\text{im}(d_1) = 2\mathbb{Z}_{(2)},$$

we have

$$\text{im}(d_1) = \ker(d_0).$$

So the localized sequence remains exact.

3.8 8. The main counterexample: tensoring with a quotient can destroy exactness

Start with the exact sequence over $\mathbb{Z}$:

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\cdot 2} \mathbb{Z} \longrightarrow \mathbb{Z}/2\mathbb{Z} \longrightarrow 0.$$

Now tensor the whole sequence with $\mathbb{Z}/2\mathbb{Z}$. We obtain

$$0 \longrightarrow \mathbb{Z}/2\mathbb{Z} \xrightarrow{d'_1} \mathbb{Z}/2\mathbb{Z} \xrightarrow{d'_0} (\mathbb{Z}/2\mathbb{Z}) \otimes_{\mathbb{Z}} (\mathbb{Z}/2\mathbb{Z}) \longrightarrow 0.$$

Let us identify the maps.

Step 1: the map d'_1

This is induced by multiplication by 2. But modulo 2, multiplication by 2 is zero. So

$$d'_1 = 0.$$

Hence

$$\operatorname{im}(d'_1) = 0, \quad \operatorname{ker}(d'_1) = \mathbb{Z}/2\mathbb{Z}.$$

Step 2: the map d'_0

The tensor product

$$(\mathbb{Z}/2\mathbb{Z}) \otimes_{\mathbb{Z}} (\mathbb{Z}/2\mathbb{Z})$$

is isomorphic to $\mathbb{Z}/2\mathbb{Z}$. So d'_0 is a map

$$d'_0 : \mathbb{Z}/2\mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z}.$$

In fact this map is an isomorphism in this case, so

$$\operatorname{ker}(d'_0) = 0 \quad \text{and} \quad \operatorname{im}(d'_0) = \mathbb{Z}/2\mathbb{Z}.$$

Where exactness fails

At the middle term $\mathbb{Z}/2\mathbb{Z}$, exactness would require

$$\operatorname{im}(d'_1) = \operatorname{ker}(d'_0).$$

Here both are 0, so exactness at the middle term still holds.

But at the first nonzero term, exactness would require

$$\operatorname{ker}(d'_1) = 0.$$

Instead,

$$\operatorname{ker}(d'_1) = \mathbb{Z}/2\mathbb{Z} \neq 0.$$

So the complex is no longer exact at the left.

Conclusion

Tensoring an exact sequence with a quotient ring can destroy exactness on the left. This is the basic phenomenon measured by Tor.

3.9 9. A field example: everything becomes trivial

Let K be a field and $M = K^m$. Consider the map

$$\text{id}_M : M \rightarrow M.$$

Then

$$\text{im}(\text{id}_M) = M, \quad \ker(\text{id}_M) = 0.$$

So

$$0 \longrightarrow M \xrightarrow{\text{id}} M \longrightarrow 0$$

is exact.

This is why free resolutions over a field are usually trivial: every module is already free.

3.10 10. Final summary of the pattern

In all of the exact examples above, the key equalities are:

$$\text{im}(d_{i+1}) = \ker(d_i).$$

More concretely:

- For

$$0 \rightarrow \mathbb{Z} \xrightarrow{n} \mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z} \rightarrow 0,$$

we have

$$\text{im}(\cdot n) = n\mathbb{Z} = \ker(\mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}).$$

- For

$$0 \rightarrow k[x] \xrightarrow{f} k[x] \rightarrow k[x]/(f) \rightarrow 0,$$

we have

$$\text{im}(\cdot f) = (f) = \ker(k[x] \rightarrow k[x]/(f)).$$

- For

$$0 \rightarrow A \xrightarrow{\begin{pmatrix} -y \\ x \end{pmatrix}} A^2 \xrightarrow{(x \ y)} A \rightarrow k \rightarrow 0,$$

we have

$$\text{im}(A \rightarrow A^2) = \ker(A^2 \rightarrow A),$$

and

$$\text{im}(A^2 \rightarrow A) = (x, y) = \ker(A \rightarrow k).$$

- For localization, exactness survives because localization preserves kernels and images in the correct way.
- For quotienting by tensor product, exactness can fail because tensor product is not left exact.

So when checking a proposed free resolution, the practical method is always the same:

write the maps explicitly $\rightarrow$ compute images $\rightarrow$ compute kernels $\rightarrow$ compare them.

4 More examples and counterexamples: maps, images, kernels, exactness, and failures

Purpose. In this section we continue the previous discussion with more examples and counterexamples. For each complex or short exact sequence, we write the maps explicitly and compute their images and kernels step by step. We also include examples showing that a module may fail to be free, may fail to be projective, and that a short exact sequence may fail to split.

4.1 1. The module $A/(x^2)$ over $A = k[x]$

Consider the sequence

$$0 \longrightarrow A \xrightarrow{d_1} A \xrightarrow{d_0} A/(x^2) \longrightarrow 0,$$

where

$$d_1(f) = x^2 f \quad \text{and} \quad d_0(g) = g + (x^2).$$

Image and kernel of d_1

The map d_1 is multiplication by x^2 , so

$$\text{im}(d_1) = x^2 A = (x^2).$$

Its kernel is

$$\ker(d_1) = \{f \in A : x^2 f = 0\}.$$

Since $A = k[x]$ is an integral domain and $x^2 \neq 0$, this gives

$$\ker(d_1) = 0.$$

Image and kernel of d_0

The map d_0 is the quotient map, so it is surjective:

$$\text{im}(d_0) = A/(x^2).$$

Its kernel is

$$\ker(d_0) = \{g \in A : g + (x^2) = (x^2)\} = (x^2).$$

Thus

$$\text{im}(d_1) = \ker(d_0).$$

Conclusion

The sequence

$$0 \longrightarrow k[x] \xrightarrow{\cdot x^2} k[x] \longrightarrow k[x]/(x^2) \longrightarrow 0$$

is exact.

Remark. This looks formally similar to the case $A/(x)$, but the quotient is different: in $A/(x^2)$, the class of x is nonzero but nilpotent, since

$$x^2 = 0 \quad \text{in } A/(x^2).$$

4.2 2. The module $A/(xy)$ over $A = k[x, y]$

Now let

$$A = k[x, y].$$

Consider

$$0 \longrightarrow A \xrightarrow{d_1} A \xrightarrow{d_0} A/(xy) \longrightarrow 0,$$

where

$$d_1(f) = xyf \quad \text{and} \quad d_0(g) = g + (xy).$$

Image and kernel of d_1

The image is

$$\text{im}(d_1) = xyA = (xy).$$

The kernel is

$$\ker(d_1) = \{f \in A : xyf = 0\}.$$

Since $A = k[x, y]$ is an integral domain and $xy \neq 0$, we get

$$\ker(d_1) = 0.$$

Image and kernel of d_0

The map d_0 is surjective, so

$$\text{im}(d_0) = A/(xy).$$

Its kernel is

$$\ker(d_0) = (xy).$$

Hence

$$\text{im}(d_1) = \ker(d_0).$$

Conclusion

The sequence

$$0 \longrightarrow k[x, y] \xrightarrow{\cdot xy} k[x, y] \longrightarrow k[x, y]/(xy) \longrightarrow 0$$

is exact.

Remark. The quotient $A/(xy)$ is not a domain, because in it one has

$$\bar{x}\bar{y} = 0$$

while neither $\bar{x}$ nor $\bar{y}$ is zero. So exactness of the sequence does not require the quotient itself to be a domain.

4.3 3. The module $A/(x)$ over $A = k[x, y]$

Let

$$A = k[x, y].$$

Consider

$$0 \longrightarrow A \xrightarrow{d_1} A \xrightarrow{d_0} A/(x) \longrightarrow 0,$$

where

$$d_1(f) = xf \quad \text{and} \quad d_0(g) = g + (x).$$

Image and kernel of d_1

We have

$$\text{im}(d_1) = xA = (x).$$

Also

$$\ker(d_1) = \{f \in A : xf = 0\} = 0,$$

because A is a domain.

Image and kernel of d_0

The quotient map is surjective, so

$$\text{im}(d_0) = A/(x).$$

Its kernel is

$$\ker(d_0) = (x).$$

Hence

$$\text{im}(d_1) = \ker(d_0).$$

Conclusion

The sequence

$$0 \longrightarrow k[x, y] \xrightarrow{\cdot x} k[x, y] \longrightarrow k[x, y]/(x) \longrightarrow 0$$

is exact.

Remark. There is a natural isomorphism

$$k[x, y]/(x) \cong k[y].$$

So this is a free resolution of $k[y]$ as a $k[x, y]$ -module.

4.4 4. The module $A/(x, y^2)$ over $A = k[x, y]$

Now consider a slightly more interesting quotient:

$$A = k[x, y], \quad M = A/(x, y^2).$$

A free resolution is

$$0 \longrightarrow A \xrightarrow{d_2} A^2 \xrightarrow{d_1} A \xrightarrow{d_0} A/(x, y^2) \longrightarrow 0,$$

where

$$d_2(c) = (-cy^2, cx),$$

$$d_1(a, b) = ax + by^2,$$

and

$$d_0(f) = f + (x, y^2).$$

Image and kernel of d_0

The map d_0 is the quotient map, so

$$\text{im}(d_0) = A/(x, y^2).$$

Its kernel is

$$\ker(d_0) = (x, y^2).$$

Image and kernel of d_1

The image is

$$\text{im}(d_1) = \{ax + by^2 : a, b \in A\} = (x, y^2).$$

Hence

$$\text{im}(d_1) = \ker(d_0).$$

Now compute $\ker(d_1)$. Suppose $(a, b) \in A^2$ satisfies

$$d_1(a, b) = ax + by^2 = 0.$$

Then

$$ax = -by^2.$$

So x divides by^2 . Since x and y^2 are relatively prime in the UFD $k[x, y]$, it follows that $x \mid b$. Write

$$b = xc$$

for some $c \in A$. Substituting back gives

$$ax + xcy^2 = 0,$$

so

$$x(a + cy^2) = 0.$$

Because A is a domain, this implies

$$a = -cy^2.$$

Therefore

$$(a, b) = (-cy^2, cx) = d_2(c).$$

So

$$\ker(d_1) = \operatorname{im}(d_2).$$

Kernel of d_2

We have

$$\ker(d_2) = \{c \in A : (-cy^2, cx) = (0, 0)\}.$$

Thus

$$cy^2 = 0 \quad \text{and} \quad cx = 0.$$

Since A is a domain, we get

$$c = 0.$$

Hence

$$\ker(d_2) = 0.$$

Conclusion

The sequence

$$0 \longrightarrow A \xrightarrow{\begin{pmatrix} -y^2 \\ x \end{pmatrix}} A^2 \xrightarrow{\begin{pmatrix} x & y^2 \end{pmatrix}} A \longrightarrow A/(x, y^2) \longrightarrow 0$$

is exact.

Remark. This is the same pattern as the standard resolution of $A/(x, y)$, but with y replaced by y^2 .

4.5 5. The ideal (x, y) as a module over $A = k[x, y]$

This is an important counterexample-type example. Let

$$I = (x, y) \subseteq A = k[x, y].$$

Consider the sequence

$$0 \longrightarrow A \xrightarrow{d_1} A^2 \xrightarrow{d_0} I \longrightarrow 0,$$

where

$$d_1(c) = (-cy, cx),$$

and

$$d_0(a, b) = ax + by.$$

Image and kernel of d_0

By definition of the ideal (x, y) , every element of I has the form

$$ax + by$$

for some $a, b \in A$. So d_0 is surjective:

$$\text{im}(d_0) = I.$$

Now compute its kernel. Suppose

$$d_0(a, b) = ax + by = 0.$$

Then

$$ax = -by.$$

As in the earlier calculation, $y \mid a$. So write

$$a = -cy.$$

Then

$$(-cy)x + by = 0$$

implies

$$b = cx.$$

Thus

$$(a, b) = (-cy, cx) = d_1(c).$$

So

$$\ker(d_0) = \text{im}(d_1).$$

Kernel of d_1

We have

$$\ker(d_1) = \{c \in A : (-cy, cx) = (0, 0)\}.$$

Hence

$$cy = 0 \quad \text{and} \quad cx = 0.$$

Since A is a domain,

$$c = 0.$$

So

$$\ker(d_1) = 0.$$

Conclusion

The sequence

$$0 \longrightarrow A \xrightarrow{\begin{pmatrix} -y \\ x \end{pmatrix}} A^2 \xrightarrow{(x \ y)} (x, y) \longrightarrow 0$$

is exact.

Why this is a useful counterexample. The module (x, y) has a short free resolution, but (x, y) itself is *not* free.

Indeed, if (x, y) were free of rank 1, then as an ideal it would be principal, which it is not. If it were free of rank $r \geq 2$, then after tensoring with the fraction field of A , its rank would be at least 2, but as a nonzero ideal inside a domain it has rank 1. So (x, y) is not free.

Thus this is a good counterexample to the false statement:

“a module with a short free resolution must itself be free.”

4.6 6. Localization at (x) : the module $k[x]_{(x)}/(x)$

Let

$$A = k[x], \quad S = A \setminus (x), \quad A_{(x)} = S^{-1}A.$$

Localizing the standard exact sequence

$$0 \longrightarrow A \xrightarrow{x} A \longrightarrow A/(x) \longrightarrow 0$$

gives

$$0 \longrightarrow A_{(x)} \xrightarrow{d_1} A_{(x)} \xrightarrow{d_0} A_{(x)}/xA_{(x)} \longrightarrow 0,$$

where

$$d_1(r) = xr.$$

Image and kernel of d_1

The image is

$$\text{im}(d_1) = xA_{(x)}.$$

The kernel is

$$\ker(d_1) = \{r \in A_{(x)} : xr = 0\} = 0,$$

because $A_{(x)}$ is still a domain.

Image and kernel of d_0

The quotient map is surjective:

$$\text{im}(d_0) = A_{(x)}/xA_{(x)}.$$

Its kernel is

$$\ker(d_0) = xA_{(x)}.$$

Thus

$$\text{im}(d_1) = \ker(d_0).$$

Conclusion

The sequence

$$0 \longrightarrow k[x]_{(x)} \xrightarrow{\cdot x} k[x]_{(x)} \longrightarrow k[x]_{(x)}/(x) \longrightarrow 0$$

is exact.

Moreover,

$$k[x]_{(x)}/(x) \cong k.$$

So this is the local version of the basic resolution of the residue field at $x = 0$.

4.7 7. Localization at $(x - 1)$: the module $A/(x)$ disappears

Let again $A = k[x]$, but now localize at the prime ideal $(x - 1)$:

$$A_{(x-1)}.$$

Start with

$$0 \longrightarrow A \xrightarrow{\cdot x} A \longrightarrow A/(x) \longrightarrow 0.$$

After localizing, we obtain

$$0 \longrightarrow A_{(x-1)} \xrightarrow{d_1} A_{(x-1)} \xrightarrow{d_0} (A/(x))_{(x-1)} \longrightarrow 0.$$

Here $x \notin (x - 1)$, so x becomes a unit in $A_{(x-1)}$. Therefore multiplication by x is an isomorphism:

$$\text{im}(d_1) = A_{(x-1)}, \quad \ker(d_1) = 0.$$

Since d_1 is surjective, the quotient is zero:

$$(A/(x))_{(x-1)} = 0.$$

Then d_0 is simply the zero map

$$d_0 : A_{(x-1)} \rightarrow 0,$$

so

$$\operatorname{im}(d_0) = 0, \quad \ker(d_0) = A_{(x-1)}.$$

Thus exactness holds:

$$\operatorname{im}(d_1) = A_{(x-1)} = \ker(d_0).$$

Interpretation. Localization at $(x - 1)$ does not see the module $A/(x)$, because x becomes invertible there.

4.8 8. A counterexample: $A/(x)$ is not free over $A = k[x]$

Let

$$A = k[x], \quad M = A/(x).$$

We already have the exact sequence

$$0 \longrightarrow A \xrightarrow{\cdot x} A \longrightarrow M \longrightarrow 0.$$

Now we show that M is not free as an A -module.

Reason

In M , the class $\bar{1}$ is nonzero, but

$$x \cdot \bar{1} = 0.$$

So M has a nonzero element annihilated by the nonzero element $x \in A$.

But a free module over the domain $A = k[x]$ is torsion-free. Indeed, if

$$F \cong A^{(I)}$$

and $0 \neq a \in A$, then $af = 0$ implies $f = 0$ coordinatewise.

Therefore $M = A/(x)$ cannot be free.

4.9 9. A counterexample: $A/(x)$ is not projective over $A = k[x]$

We now show that $M = A/(x)$ is not projective.

If M were projective, then the exact sequence

$$0 \longrightarrow A \xrightarrow{\cdot x} A \xrightarrow{\pi} M \longrightarrow 0$$

would split. So there would exist an A -linear map

$$s : M \rightarrow A$$

such that

$$\pi \circ s = \text{id}_M.$$

Let

$$a = s(\bar{1}) \in A.$$

Because $\pi(a) = \bar{1}$, we must have

$$a \equiv 1 \pmod{(x)}.$$

So $a \neq 0$.

But s is A -linear, and in M we have

$$x \cdot \bar{1} = 0.$$

Hence

$$xa = s(x \cdot \bar{1}) = s(0) = 0.$$

Since $A = k[x]$ is a domain and $x \neq 0$, this forces

$$a = 0,$$

a contradiction.

Therefore no such section s exists, and M is not projective.

4.10 10. A counterexample: the short exact sequence does not split

The same example gives a nonsplit short exact sequence:

$$0 \longrightarrow A \xrightarrow{x} A \longrightarrow A/(x) \longrightarrow 0.$$

We already proved above that $A/(x)$ is not projective, so this sequence does not split.

Equivalently, there is no A -module isomorphism

$$A \cong A \oplus A/(x)$$

compatible with this sequence.

Thus this is a concrete counterexample to the false statement:

“every short exact sequence of modules splits.”

4.11 11. Another localization example: $A/(xy)$ at (x)

Let

$$A = k[x, y],$$

and consider the exact sequence

$$0 \longrightarrow A \xrightarrow{\cdot xy} A \longrightarrow A/(xy) \longrightarrow 0.$$

Now localize at the prime ideal (x) . In $A_{(x)}$, the element y is a unit, because $y \notin (x)$. So multiplication by xy has the same image as multiplication by x :

$$xyA_{(x)} = xA_{(x)}.$$

Hence the localized exact sequence is

$$0 \longrightarrow A_{(x)} \xrightarrow{\cdot xy} A_{(x)} \longrightarrow (A/(xy))_{(x)} \longrightarrow 0,$$

and the quotient is isomorphic to

$$(A/(xy))_{(x)} \cong A_{(x)}/xA_{(x)}.$$

Its image and kernel behave exactly as in the basic principal case:

$$\text{im}(\cdot xy) = xA_{(x)}, \quad \ker(\cdot xy) = 0,$$

and the quotient map has kernel $xA_{(x)}$.

So near the prime (x) , the module $A/(xy)$ looks just like $A/(x)$.

4.12 12. Summary of lessons from these examples

These examples illustrate several important points.

1. Principal quotients such as

$$A/(x^2), \quad A/(xy), \quad A/(x)$$

have short free resolutions of length 1.

2. Quotients by two generators, such as

$$A/(x, y^2),$$

typically require one more step, because one must also record a relation among the relations.

3. An ideal such as (x, y) may have a short free resolution without being free itself.
4. Localization preserves exactness and often simplifies a module by turning some elements into units.

5. The module $A/(x)$ over $A = k[x]$ is a concrete example of a module that is

- not free,
- not projective,
- and gives a short exact sequence that does not split.

So the practical method remains:

write the maps explicitly $\longrightarrow$ compute images $\longrightarrow$ compute kernels $\longrightarrow$ compare them.

That is the basic mechanism behind checking exactness and understanding resolutions in concrete algebra.

5 Minimal free resolutions over polynomial rings, local rings, and localizations

Purpose. In the previous sections, we studied free resolutions and checked exactness by computing maps, images, and kernels explicitly. We now refine that discussion by introducing *minimal* free resolutions. A minimal free resolution is a free resolution with no redundant free summands. This notion is especially clean over local rings and graded polynomial rings.

5.1 1. What does minimal mean?

Let

$$\cdots \longrightarrow F_2 \xrightarrow{d_2} F_1 \xrightarrow{d_1} F_0 \longrightarrow M \longrightarrow 0$$

be a free resolution of an R -module M .

Intuitively, this free resolution is *minimal* if no free direct summand can be cancelled from two consecutive terms. In other words, the resolution contains no unnecessary part.

Local criterion

Suppose $(R, \mathfrak{m})$ is a local ring. Then a free resolution is minimal if and only if

$$d_i(F_i) \subseteq \mathfrak{m}F_{i-1} \quad \text{for all } i \geq 1.$$

Equivalently, if one writes the maps d_i as matrices with respect to chosen bases, then every matrix entry lies in the maximal ideal $\mathfrak{m}$.

So over a local ring, minimality means:

no entry of any differential matrix is a unit.

Residue-field criterion

Still over a local ring $(R, \mathfrak{m})$, let

$$k = R/\mathfrak{m}$$

be the residue field. Then a free resolution is minimal if and only if, after tensoring with k , all differentials become zero:

$$d_i \otimes_R k = 0 \quad \text{for all } i.$$

This is equivalent to the previous criterion, because an element of R becomes zero in k exactly when it lies in $\mathfrak{m}$.

Graded criterion for polynomial rings

Now let

$$S = k[x_1, \dots, x_n]$$

be a polynomial ring with its standard grading. A graded free resolution is minimal if and only if every entry in every differential matrix has positive degree, or equivalently lies in the homogeneous maximal ideal

$$(x_1, \dots, x_n).$$

So in graded polynomial rings, minimality means:

$$\text{no matrix entry is a nonzero constant.}$$

5.2 2. The first example: k over $k[x]$

Let

$$S = k[x], \quad M = S/(x) \cong k.$$

A free resolution is

$$0 \longrightarrow S \xrightarrow{x} S \longrightarrow k \longrightarrow 0.$$

Why this is exact

The map $S \rightarrow k$ is the quotient map modulo (x) , so its kernel is

$$(x).$$

The map $S \xrightarrow{x} S$ has image

$$xS = (x),$$

and because S is a domain, its kernel is zero. Thus the sequence is exact.

Why this is minimal

The only differential is multiplication by x , and

$$x \in (x).$$

So the matrix entry lies in the homogeneous maximal ideal. Therefore the resolution is minimal.

Graded form

In graded notation one usually writes

$$0 \longrightarrow S(-1) \xrightarrow{x} S \longrightarrow k \longrightarrow 0.$$

The shift $S(-1)$ is included so that the map has degree 0.

5.3 3. The local version: k over $k[x]_{(x)}$

Let

$$R = k[x]_{(x)}, \quad \mathfrak{m} = (x)R.$$

Then the residue field is

$$R/\mathfrak{m} \cong k.$$

A free resolution is

$$0 \longrightarrow R \xrightarrow{x} R \longrightarrow k \longrightarrow 0.$$

Why this is exact

The quotient map $R \rightarrow R/(x)$ has kernel $(x)R$, and multiplication by x has image $(x)R$. Also R is a domain, so multiplication by x has kernel zero. Hence the sequence is exact.

Why this is minimal

The differential is multiplication by x , and

$$x \in \mathfrak{m}.$$

So

$$d_1(R) \subseteq \mathfrak{m}R,$$

which is exactly the local criterion for minimality.

5.4 4. The same picture at another point

Let

$$R = k[x]_{(x-a)}, \quad \mathfrak{m} = (x-a)R.$$

Then

$$R/\mathfrak{m} \cong k.$$

A free resolution of the residue field is

$$0 \longrightarrow R \xrightarrow{\cdot(x-a)} R \longrightarrow k \longrightarrow 0.$$

This is exact for the same reason as before, and it is minimal because

$$x-a \in \mathfrak{m}.$$

So the residue field at any closed point has the same kind of minimal free resolution over the corresponding local ring.

5.5 5. The residue field k over $k[x, y]$

Let

$$S = k[x, y], \quad M = S/(x, y) \cong k.$$

The standard free resolution is

$$0 \longrightarrow S \xrightarrow{d_2} S^2 \xrightarrow{d_1} S \longrightarrow k \longrightarrow 0,$$

where

$$d_2(c) = (-cy, cx), \quad d_1(a, b) = ax + by.$$

In matrix form,

$$0 \longrightarrow S \xrightarrow{\begin{pmatrix} -y \\ x \end{pmatrix}} S^2 \xrightarrow{\begin{pmatrix} x & y \end{pmatrix}} S \longrightarrow k \longrightarrow 0.$$

Why this is exact

The map $S \rightarrow k$ is evaluation at $(0, 0)$, so its kernel is (x, y) . The image of d_1 is also (x, y) , since

$$\text{im}(d_1) = \{ax + by : a, b \in S\} = (x, y).$$

Now suppose $(a, b) \in \ker(d_1)$. Then

$$ax + by = 0.$$

As in the earlier exactness check, this implies

$$(a, b) = (-cy, cx)$$

for some $c \in S$, so

$$\ker(d_1) = \operatorname{im}(d_2).$$

Finally, since S is a domain,

$$d_2(c) = (-cy, cx) = (0, 0)$$

implies $c = 0$. Thus the sequence is exact.

Why this is minimal

The differential matrices contain only x and y , and both lie in

$$(x, y).$$

So there are no unit entries and no nonzero constants. Hence this is the minimal graded free resolution of k over $k[x, y]$.

Graded form

In graded notation one writes

$$0 \longrightarrow S(-2) \xrightarrow{\begin{pmatrix} -y \\ x \end{pmatrix}} S(-1)^2 \xrightarrow{\begin{pmatrix} x & y \end{pmatrix}} S \longrightarrow k \longrightarrow 0.$$

5.6 6. The local version over $k[x, y]_{(x, y)}$

Let

$$R = k[x, y]_{(x, y)}, \quad \mathfrak{m} = (x, y)R.$$

Then the residue field is

$$R/\mathfrak{m} \cong k.$$

A free resolution is

$$0 \longrightarrow R \xrightarrow{\begin{pmatrix} -y \\ x \end{pmatrix}} R^2 \xrightarrow{\begin{pmatrix} x & y \end{pmatrix}} R \longrightarrow k \longrightarrow 0.$$

This is exact for the same reason as over $k[x, y]$, and it is minimal because every entry in the differential matrices lies in $\mathfrak{m}$.

Thus this is the minimal free resolution of the residue field over the local ring $k[x, y]_{(x, y)}$.

5.7 7. The module $S/(x, y^2)$

Let

$$S = k[x, y], \quad M = S/(x, y^2).$$

A free resolution is

$$0 \longrightarrow S \xrightarrow{d_2} S^2 \xrightarrow{d_1} S \longrightarrow M \longrightarrow 0,$$

where

$$d_2(c) = (-cy^2, cx), \quad d_1(a, b) = ax + by^2.$$

In matrix form,

$$0 \longrightarrow S \xrightarrow{\begin{pmatrix} -y^2 \\ x \end{pmatrix}} S^2 \xrightarrow{\begin{pmatrix} x & y^2 \end{pmatrix}} S \longrightarrow S/(x, y^2) \longrightarrow 0.$$

Why this is exact

The quotient map $S \rightarrow S/(x, y^2)$ has kernel

$$(x, y^2).$$

The image of d_1 is

$$\text{im}(d_1) = \{ax + by^2 : a, b \in S\} = (x, y^2),$$

so exactness holds at S .

Now let $(a, b) \in \ker(d_1)$. Then

$$ax + by^2 = 0.$$

As in the earlier computation, this implies

$$(a, b) = (-cy^2, cx)$$

for some $c \in S$. Thus

$$\ker(d_1) = \text{im}(d_2).$$

Finally,

$$d_2(c) = (-cy^2, cx) = (0, 0)$$

forces $c = 0$ because S is a domain. So the sequence is exact.

Why this is minimal

The differential entries are x and y^2 , both of which lie in

$$(x, y)$$

and both have positive degree. Hence the resolution is minimal.

Graded form

Since the generators x and y^2 have different degrees, the graded version is

$$0 \longrightarrow S(-3) \xrightarrow{\begin{pmatrix} -y^2 \\ x \end{pmatrix}} S(-1) \oplus S(-2) \xrightarrow{\begin{pmatrix} x & y^2 \end{pmatrix}} S \longrightarrow S/(x, y^2) \longrightarrow 0.$$

5.8 8. The ideal (x, y) and its minimal free resolution

Let

$$S = k[x, y], \quad I = (x, y).$$

A free resolution of I is

$$0 \longrightarrow S \xrightarrow{d_1} S^2 \xrightarrow{d_0} I \longrightarrow 0,$$

where

$$d_1(c) = (-cy, cx), \quad d_0(a, b) = ax + by.$$

In matrix form,

$$0 \longrightarrow S \xrightarrow{\begin{pmatrix} -y \\ x \end{pmatrix}} S^2 \xrightarrow{\begin{pmatrix} x & y \end{pmatrix}} I \longrightarrow 0.$$

Why this is exact

The map d_0 is surjective by definition of the ideal (x, y) . Its kernel consists of pairs (a, b) satisfying

$$ax + by = 0,$$

and as before one checks that every such pair has the form

$$(a, b) = (-cy, cx) = d_1(c).$$

So

$$\ker(d_0) = \operatorname{im}(d_1).$$

Finally, since S is a domain, $\ker(d_1) = 0$.

Thus the sequence is exact.

Why this is minimal

Again the differential entries are x and y , both lying in (x, y) , so the resolution is minimal.

Important point. The module $I = (x, y)$ is not free, but its free resolution is minimal. So a minimal free resolution does not mean that the module itself is free. It means only that the chosen free modules in the resolution are as small as possible.

5.9 9. A nonminimal resolution: a redundant free summand

To understand minimality better, it is useful to see a free resolution that is exact but not minimal.

Let

$$R = k[x]_{(x)}, \quad \mathfrak{m} = (x).$$

The minimal free resolution of the residue field k is

$$0 \longrightarrow R \xrightarrow{\cdot x} R \longrightarrow k \longrightarrow 0.$$

Now consider instead

$$0 \longrightarrow R \xrightarrow{d_2} R^2 \xrightarrow{d_1} R \longrightarrow k \longrightarrow 0,$$

where

$$d_2(a) = (a, xa), \quad d_1(u, v) = -xu + v.$$

In matrix form,

$$d_2 = \begin{pmatrix} 1 \\ x \end{pmatrix}, \quad d_1 = (-x \ 1).$$

Why this is exact

First,

$$d_1(d_2(a)) = d_1(a, xa) = -xa + xa = 0,$$

so $\text{im}(d_2) \subseteq \ker(d_1)$.

Now let $(u, v) \in \ker(d_1)$. Then

$$-xu + v = 0,$$

so

$$v = xu.$$

Hence

$$(u, v) = (u, xu) = d_2(u),$$

which shows

$$\ker(d_1) = \text{im}(d_2).$$

Also,

$$d_2(a) = (a, xa) = (0, 0)$$

implies $a = 0$, so $\ker(d_2) = 0$.

Therefore the sequence is exact.

Why this is not minimal

The matrices of d_2 and d_1 contain the unit 1, and

$$1 \notin \mathfrak{m} = (x).$$

So the local minimality criterion fails.

Hence this exact free resolution is not minimal.

Interpretation. The unit entry reveals a redundant free summand that can be cancelled. After cancellation, one returns to the minimal free resolution

$$0 \longrightarrow R \xrightarrow{x} R \longrightarrow k \longrightarrow 0.$$

5.10 10. The residue-field test for minimality

We now illustrate the criterion involving tensoring with the residue field.

Let $(R, \mathfrak{m})$ be a local ring and let

$$k = R/\mathfrak{m}.$$

Minimal example

Take the minimal free resolution

$$0 \longrightarrow R \xrightarrow{\cdot x} R \longrightarrow k \longrightarrow 0$$

over $R = k[x]_{(x)}$.

Tensoring with k , the map multiplication by x becomes zero, because $x \in \mathfrak{m}$ and hence $x = 0$ in k . So after tensoring, the differential is the zero map. This agrees with the minimality criterion.

Nonminimal example

Now take the nonminimal resolution

$$0 \longrightarrow R \xrightarrow{\begin{pmatrix} 1 \\ x \end{pmatrix}} R^2 \xrightarrow{(-x \ 1)} R \longrightarrow k \longrightarrow 0.$$

Tensoring with k , the entries x become 0, but the entries 1 remain 1. So the differentials become

$$k \xrightarrow{\begin{pmatrix} 1 \\ 0 \end{pmatrix}} k^2 \xrightarrow{(0 \ 1)} k,$$

which are not zero maps.

Therefore the resolution is not minimal.

5.11 11. Betti numbers and why minimal resolutions matter

One of the main reasons minimal resolutions are important is that their free ranks are intrinsic invariants.

If

$$\cdots \longrightarrow F_2 \longrightarrow F_1 \longrightarrow F_0 \longrightarrow M \longrightarrow 0$$

is a minimal free resolution, then the ranks of the F_i are called the *Betti numbers* of M over the ring.

Example: k over $k[x]$

The minimal graded free resolution is

$$0 \longrightarrow S(-1) \longrightarrow S \longrightarrow k \longrightarrow 0.$$

So the free ranks are

$$1, 1.$$

Example: k over $k[x, y]$

The minimal graded free resolution is

$$0 \longrightarrow S(-2) \longrightarrow S(-1)^2 \longrightarrow S \longrightarrow k \longrightarrow 0.$$

So the free ranks are

$$1, 2, 1.$$

These numbers do not depend on the choice of minimal free resolution.

5.12 12. Final summary

Minimal free resolutions sharpen the theory of free resolutions by removing all redundant free summands.

Over a local ring $(R, \mathfrak{m})$, minimality means:

$$d_i(F_i) \subseteq \mathfrak{m}F_{i-1} \quad \text{for all } i,$$

or equivalently, all differentials become zero after tensoring with the residue field $R/\mathfrak{m}$.

Over a graded polynomial ring $k[x_1, \dots, x_n]$, minimality means that all differential entries have positive degree, or equivalently lie in the homogeneous maximal ideal $(x_1, \dots, x_n)$.

The main examples are:

- k over $k[x]$: minimal free resolution of length 1,
- k over $k[x, y]$: minimal free resolution of length 2,
- k over localizations such as $k[x]_{(x)}$ and $k[x, y]_{(x, y)}$,
- $S/(x, y^2)$: a minimal two-step resolution,
- (x, y) : a module with a minimal free resolution, even though the module itself is not free.

A nonminimal free resolution is detected by the appearance of a unit in one of the differential matrices. Such a unit signals a cancellable summand. So minimal free resolutions are the most efficient free resolutions and provide the correct intrinsic homological data of the module.

6 Syzygies, projective dimension, and Betti numbers: concrete examples over $\mathbb{Z}$, $k[x]$, $k[x, y]$, quotients, and localizations

Purpose. After studying free resolutions, exactness, and minimal free resolutions, the next natural step is to name and organize the structure that was already appearing in our kernel computations. The kernels in a free resolution are called *syzygy modules*. The length of a shortest projective or free resolution gives the *projective dimension*. And in a minimal free resolution, the ranks of the free modules are the *Betti numbers*. This section develops these notions through the same concrete examples used earlier.

6.1 1. From kernels to syzygies

Let R be a ring and let

$$\cdots \longrightarrow F_2 \xrightarrow{d_2} F_1 \xrightarrow{d_1} F_0 \xrightarrow{\varepsilon} M \longrightarrow 0$$

be a free resolution of an R -module M .

The first kernel

$$\ker(\varepsilon) \subseteq F_0$$

is the module of relations among the chosen generators of M . It is called the *first syzygy module* of M , and is often denoted

$$\Omega^1(M).$$

More generally, for $i \geq 1$, the module

$$\Omega^i(M) = \ker(d_{i-1})$$

is called the i -th syzygy module of M .

So:

- $\Omega^1(M)$ is the module of relations among generators,
- $\Omega^2(M)$ is the module of relations among relations,
- $\Omega^3(M)$ is the module of relations among relations among relations,
- and so on.

Thus a free resolution can be read as a chain of syzygies:

$$\Omega^1(M), \Omega^2(M), \Omega^3(M), \dots$$

6.2 2. First example over $\mathbb{Z}$: the module $\mathbb{Z}/n\mathbb{Z}$

Consider the exact sequence

$$0 \longrightarrow \mathbb{Z} \xrightarrow{n} \mathbb{Z} \longrightarrow \mathbb{Z}/n\mathbb{Z} \longrightarrow 0.$$

The first syzygy

The map $\mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$ is the quotient map. Its kernel is

$$\ker(\mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}) = n\mathbb{Z}.$$

So

$$\Omega^1(\mathbb{Z}/n\mathbb{Z}) = n\mathbb{Z}.$$

Since multiplication by n identifies $\mathbb{Z}$ with $n\mathbb{Z}$, we have

$$\Omega^1(\mathbb{Z}/n\mathbb{Z}) \cong \mathbb{Z}.$$

The second syzygy

The map

$$\mathbb{Z} \xrightarrow{\cdot n} \mathbb{Z}$$

has kernel

$$\ker(\cdot n) = 0.$$

So

$$\Omega^2(\mathbb{Z}/n\mathbb{Z}) = 0.$$

Projective dimension

Because the resolution stops after one nonzero free module on the left, the projective dimension is

$$\mathrm{pd}_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}) = 1.$$

Betti numbers

The minimal free resolution has the form

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Z}/n\mathbb{Z} \longrightarrow 0.$$

So the free ranks are

$$1, 1.$$

Thus the Betti numbers are

$$\beta_0 = 1, \quad \beta_1 = 1.$$

6.3 3. Principal quotients over $k[x]$

Let

$$S = k[x], \quad M = S/(f),$$

where $f \in S$ is nonzero. The standard free resolution is

$$0 \longrightarrow S \xrightarrow{f} S \longrightarrow S/(f) \longrightarrow 0.$$

The first syzygy

The kernel of the quotient map $S \rightarrow S/(f)$ is

$$(f).$$

Hence

$$\Omega^1(S/(f)) = (f) \cong S.$$

The second syzygy

The map $S \xrightarrow{f} S$ has kernel zero because S is a domain and $f \neq 0$. So

$$\Omega^2(S/(f)) = 0.$$

Projective dimension

Therefore

$$\text{pd}_{k[x]}(S/(f)) = 1.$$

Betti numbers

The minimal free resolution has free ranks

$$1, 1.$$

So

$$\beta_0 = 1, \quad \beta_1 = 1.$$

Special case: $k = S/(x)$

If $M = S/(x) \cong k$, then

$$0 \longrightarrow S(-1) \xrightarrow{x} S \longrightarrow k \longrightarrow 0$$

is the minimal graded free resolution. Again,

$$\beta_0 = 1, \quad \beta_1 = 1, \quad \text{pd}_{k[x]}(k) = 1.$$

6.4 4. The residue field k over $k[x, y]$

Let

$$S = k[x, y], \quad M = S/(x, y) \cong k.$$

The minimal graded free resolution is

$$0 \longrightarrow S(-2) \xrightarrow{\begin{pmatrix} -y \\ x \end{pmatrix}} S(-1)^2 \xrightarrow{\begin{pmatrix} x & y \end{pmatrix}} S \longrightarrow k \longrightarrow 0.$$

The first syzygy

The map $S \rightarrow k$ is quotient by (x, y) , so

$$\Omega^1(k) = \ker(S \rightarrow k) = (x, y).$$

Thus the first syzygy of k is the maximal ideal (x, y) .

The second syzygy

Now consider

$$d_1 : S^2 \rightarrow S, \quad d_1(a, b) = ax + by.$$

Its kernel consists of all pairs (a, b) such that

$$ax + by = 0.$$

As computed earlier, every such pair has the form

$$(a, b) = (-cy, cx)$$

for some $c \in S$. Therefore

$$\Omega^2(k) = \ker(d_1) = \{(-cy, cx) : c \in S\} \cong S.$$

The third syzygy

The map

$$d_2 : S \rightarrow S^2, \quad d_2(c) = (-cy, cx)$$

has kernel zero because S is a domain. Hence

$$\Omega^3(k) = 0.$$

Projective dimension

The minimal free resolution has length 2, so

$$\mathrm{pd}_{k[x,y]}(k) = 2.$$

Betti numbers

The free ranks are

$$1, 2, 1.$$

Thus the Betti numbers are

$$\beta_0 = 1, \quad \beta_1 = 2, \quad \beta_2 = 1.$$

Interpretation. The one generator of k gives $\beta_0 = 1$. The two relations $x = 0$ and $y = 0$ give $\beta_1 = 2$. The one relation among those relations gives $\beta_2 = 1$.

6.5 5. The module $S/(x, y^2)$

Let

$$S = k[x, y], \quad M = S/(x, y^2).$$

A minimal graded free resolution is

$$0 \longrightarrow S(-3) \xrightarrow{\begin{pmatrix} -y^2 \\ x \end{pmatrix}} S(-1) \oplus S(-2) \xrightarrow{\begin{pmatrix} x & y^2 \end{pmatrix}} S \longrightarrow M \longrightarrow 0.$$

The first syzygy

The kernel of the quotient map $S \rightarrow M$ is

$$(x, y^2).$$

So

$$\Omega^1(M) = (x, y^2).$$

The second syzygy

Now

$$d_1 : S^2 \rightarrow S, \quad d_1(a, b) = ax + by^2.$$

Its kernel consists of all pairs satisfying

$$ax + by^2 = 0.$$

As before, every such pair has the form

$$(a, b) = (-cy^2, cx)$$

for some $c \in S$. Hence

$$\Omega^2(M) \cong S.$$

The third syzygy

Since

$$d_2(c) = (-cy^2, cx)$$

has kernel zero, we get

$$\Omega^3(M) = 0.$$

Projective dimension

Therefore

$$\text{pd}_{k[x,y]}(S/(x, y^2)) = 2.$$

Betti numbers

The free ranks are again

$$1, 2, 1.$$

So

$$\beta_0 = 1, \quad \beta_1 = 2, \quad \beta_2 = 1.$$

Remark. The Betti numbers here are the same as for $k = S/(x, y)$, even though the module is different. What changes is the grading and the degrees of the generators in the minimal free resolution.

6.6 6. The ideal (x, y) as a syzygy module

Let

$$S = k[x, y], \quad I = (x, y).$$

From the minimal free resolution of k ,

$$0 \longrightarrow S(-2) \longrightarrow S(-1)^2 \longrightarrow S \longrightarrow k \longrightarrow 0,$$

we already saw that

$$\Omega^1(k) = (x, y) = I.$$

So the ideal (x, y) is itself the first syzygy module of the residue field k .

Now I has its own minimal free resolution:

$$0 \longrightarrow S(-2) \xrightarrow{\begin{pmatrix} -y \\ x \end{pmatrix}} S(-1)^2 \xrightarrow{\begin{pmatrix} x & y \end{pmatrix}} I \longrightarrow 0.$$

First syzygy of I

The kernel of

$$S^2 \rightarrow I, \quad (a, b) \mapsto ax + by$$

is

$$\{(-cy, cx) : c \in S\} \cong S.$$

So

$$\Omega^1(I) \cong S.$$

Second syzygy of I

The map

$$S \rightarrow S^2, \quad c \mapsto (-cy, cx)$$

has kernel zero. Hence

$$\Omega^2(I) = 0.$$

Projective dimension

Thus

$$\mathrm{pd}_{k[x,y]}(I) = 1.$$

Betti numbers

The minimal free resolution has free ranks

$$2, 1$$

if listed from F_0 upward as

$$F_0 = S^2, \quad F_1 = S.$$

So

$$\beta_0(I) = 2, \quad \beta_1(I) = 1.$$

Important point. The module $I = (x, y)$ is not free, but it is a first syzygy module. This is a standard phenomenon: syzygy modules are often better behaved than arbitrary modules, but they need not be free.

6.7 7. Free modules and projective dimension 0

If F is already a free R -module, then it has the trivial free resolution

$$0 \longrightarrow F \xrightarrow{\mathrm{id}} F \longrightarrow 0.$$

So

$$\mathrm{pd}_R(F) = 0.$$

In particular:

- over a field K , every module is free, so every module has projective dimension 0,
- over $\mathbb{Z}$, the module $\mathbb{Z}$ has projective dimension 0,
- over $k[x]$, the module $k[x]$ has projective dimension 0,
- over $k[x, y]$, the free module S^m has projective dimension 0.

So projective dimension measures how far a module is from already being free or projective.

6.8 8. Localizations and preservation of syzygies

Let $S \subseteq R$ be a multiplicative set. If

$$\cdots \longrightarrow F_2 \longrightarrow F_1 \longrightarrow F_0 \longrightarrow M \longrightarrow 0$$

is a free resolution of M , then localizing gives

$$\cdots \longrightarrow S^{-1}F_2 \longrightarrow S^{-1}F_1 \longrightarrow S^{-1}F_0 \longrightarrow S^{-1}M \longrightarrow 0.$$

Since localization is exact, kernels localize correctly:

$$S^{-1}(\ker d_i) \cong \ker(S^{-1}d_i).$$

Thus syzygies localize:

$$S^{-1}\Omega^i(M) \cong \Omega^i(S^{-1}M).$$

Example: $k[x]/(x)$ localized at (x)

Let

$$R = k[x], \quad M = R/(x), \quad R_{(x)} = k[x]_{(x)}.$$

The minimal free resolution over R is

$$0 \longrightarrow R \xrightarrow{\cdot x} R \longrightarrow M \longrightarrow 0.$$

After localizing at (x) , we obtain

$$0 \longrightarrow R_{(x)} \xrightarrow{\cdot x} R_{(x)} \longrightarrow M_{(x)} \longrightarrow 0.$$

The first syzygy localizes from

$$\Omega^1(M) = (x)$$

to

$$\Omega^1(M_{(x)}) = xR_{(x)}.$$

The projective dimension remains

$$\mathrm{pd}_{R_{(x)}}(M_{(x)}) = 1.$$

Example: localizing at $(x-1)$

Now localize the same module at $(x-1)$. Since x becomes a unit in $R_{(x-1)}$, we get

$$M_{(x-1)} = 0.$$

Hence the localized module has trivial resolution and

$$\mathrm{pd}_{R_{(x-1)}}(M_{(x-1)}) = 0.$$

So projective dimension can decrease after localization.

6.9 9. Quotients and syzygies

Passing to a quotient ring changes the base ring and often changes the projective dimension dramatically.

Example: $k[x]/(x)$ over two different rings

Let

$$A = k[x], \quad B = A/(x) \cong k.$$

As an A -module, B has the minimal free resolution

$$0 \longrightarrow A \xrightarrow{\cdot x} A \longrightarrow B \longrightarrow 0.$$

So

$$\mathrm{pd}_A(B) = 1.$$

But as a B -module, B is free of rank 1, so

$$\mathrm{pd}_B(B) = 0.$$

Thus projective dimension depends strongly on the chosen base ring.

Example: $\mathbb{F}_p$ over $\mathbb{Z}$ and over $\mathbb{F}_p$

As a $\mathbb{Z}$ -module,

$$0 \longrightarrow \mathbb{Z} \xrightarrow{p} \mathbb{Z} \longrightarrow \mathbb{F}_p \longrightarrow 0$$

is a minimal free resolution. Hence

$$\mathrm{pd}_{\mathbb{Z}}(\mathbb{F}_p) = 1.$$

But as an $\mathbb{F}_p$ -module, $\mathbb{F}_p$ is free, so

$$\mathrm{pd}_{\mathbb{F}_p}(\mathbb{F}_p) = 0.$$

Again, the same underlying abelian group or set can have different homological behavior over different rings.

6.10 10. Projective dimension: definition and interpretation

Let M be an R -module. The *projective dimension* of M , denoted

$$\mathrm{pd}_R(M),$$

is the shortest length of a projective resolution of M , if such a finite length exists. If no finite projective resolution exists, one writes

$$\mathrm{pd}_R(M) = \infty.$$

Since every free module is projective, one can compute projective dimension from free resolutions as well.

In our examples:

- $\text{pd}_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}) = 1,$
- $\text{pd}_{k[x]}(k[x]/(f)) = 1,$
- $\text{pd}_{k[x]}(k) = 1,$
- $\text{pd}_{k[x,y]}(k) = 2,$
- $\text{pd}_{k[x,y]}(k[x,y]/(x, y^2)) = 2,$
- $\text{pd}_{k[x,y]}((x, y)) = 1.$

So principal quotients usually have projective dimension 1, while quotients by two generators often have projective dimension 2, at least in these examples.

6.11 11. Betti numbers: numerical shadow of a minimal free resolution

Let

$$\cdots \longrightarrow F_2 \longrightarrow F_1 \longrightarrow F_0 \longrightarrow M \longrightarrow 0$$

be a minimal free resolution. If

$$F_i \cong R^{\beta_i},$$

then the integers β_i are called the *Betti numbers* of M over R .

Thus Betti numbers record how many generators are needed at each stage of a minimal free resolution.

Examples

1. For $M = \mathbb{Z}/n\mathbb{Z}$ over $\mathbb{Z}$, the minimal free resolution has ranks

$$1, 1,$$

so

$$\beta_0 = 1, \quad \beta_1 = 1.$$

2. For $M = k$ over $k[x]$, the minimal graded free resolution has ranks

$$1, 1,$$

so

$$\beta_0 = 1, \quad \beta_1 = 1.$$

3. For $M = k$ over $k[x, y]$, the minimal graded free resolution has ranks

$$1, 2, 1,$$

so

$$\beta_0 = 1, \quad \beta_1 = 2, \quad \beta_2 = 1.$$

4. For $M = k[x, y]/(x, y^2)$, the minimal graded free resolution also has ranks

$$1, 2, 1,$$

so

$$\beta_0 = 1, \quad \beta_1 = 2, \quad \beta_2 = 1.$$

5. For the ideal $I = (x, y)$, the minimal free resolution has free ranks

$$2, 1,$$

so

$$\beta_0(I) = 2, \quad \beta_1(I) = 1.$$

Graded Betti data

Over graded polynomial rings, one usually refines Betti numbers by degree shifts. For example, the minimal graded free resolution

$$0 \longrightarrow S(-2) \longrightarrow S(-1)^2 \longrightarrow S \longrightarrow k \longrightarrow 0$$

shows not only that the free ranks are 1, 2, 1, but also that the generators occur in degrees 0, 1, 2.

Similarly, for

$$S/(x, y^2),$$

the minimal graded free resolution

$$0 \longrightarrow S(-3) \longrightarrow S(-1) \oplus S(-2) \longrightarrow S \longrightarrow S/(x, y^2) \longrightarrow 0$$

shows that although the total Betti numbers are again 1, 2, 1, the degree shifts are different.

Thus Betti numbers measure the size of the resolution, while graded shifts measure where the generators occur.

6.12 12. Syzygies and the structure of the examples

It is useful to summarize the earlier examples in syzygy language.

1. For $\mathbb{Z}/n\mathbb{Z}$:

$$\Omega^1(\mathbb{Z}/n\mathbb{Z}) \cong \mathbb{Z}, \quad \Omega^2(\mathbb{Z}/n\mathbb{Z}) = 0.$$

2. For $k[x]/(f)$:

$$\Omega^1(k[x]/(f)) \cong k[x], \quad \Omega^2(k[x]/(f)) = 0.$$

3. For k over $k[x, y]$:

$$\Omega^1(k) = (x, y), \quad \Omega^2(k) \cong k[x, y], \quad \Omega^3(k) = 0.$$

4. For $k[x, y]/(x, y^2)$:

$$\Omega^1(S/(x, y^2)) = (x, y^2), \quad \Omega^2(S/(x, y^2)) \cong S, \quad \Omega^3(S/(x, y^2)) = 0.$$

5. For $I = (x, y)$:

$$\Omega^1(I) \cong S, \quad \Omega^2(I) = 0.$$

So in these concrete cases, the syzygy modules are still very explicit and computable.

6.13 13. Final perspective

The language of syzygies, projective dimension, and Betti numbers is nothing mysterious: it is simply a systematic way of organizing the kernels and ranks that already appear in explicit free resolutions.

- *Syzygies* are the kernels in the resolution.
- *Projective dimension* measures how many nontrivial steps are needed before the kernels become zero or projective.
- *Betti numbers* count the number of free generators needed at each step in a minimal free resolution.

In the examples studied here:

- principal quotients such as $\mathbb{Z}/n\mathbb{Z}$ and $k[x]/(f)$ have projective dimension 1,
- the residue field k over $k[x, y]$ has projective dimension 2,
- the module $k[x, y]/(x, y^2)$ also has projective dimension 2,
- the ideal (x, y) is itself a first syzygy module and has projective dimension 1.

Thus the next layer after free resolutions is to read those resolutions as structural invariants. This is exactly what syzygies, projective dimension, and Betti numbers do.